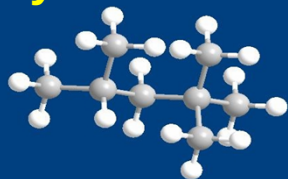
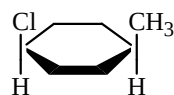


第五章 环烷烃

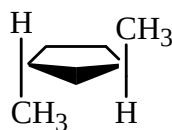
Cycloalkanes



- 顺反异构：两个相同取代基在环平面同侧为顺式，不同侧为反式



顺-1-氯-4-甲基环己烷



反-1,3-二甲基环戊烷

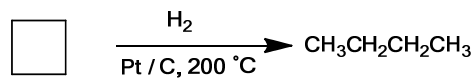
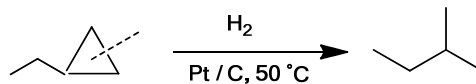
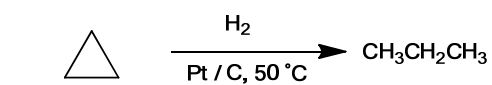
2

环烷烃的化学性质

小似烯
大似烷
高锰酸钾不破坏

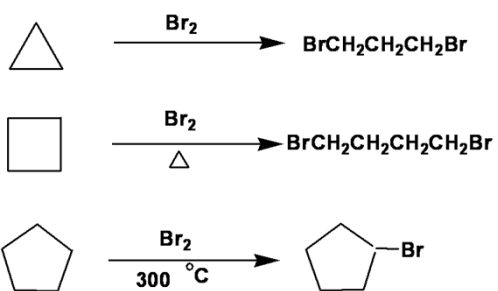
3

加氢反应—位阻最小的键开环



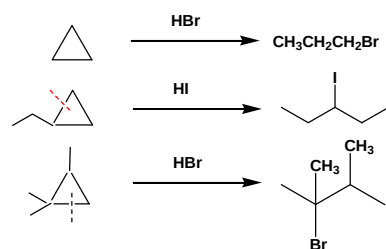
4

加卤素



5

加卤化氢



烷基取代环丙烷开环规律：环断裂发生在与氢连接最多及最少的两个成环碳上，且符合马氏规律

6

化学性质小结

大环烷烃～链形烷烃

小环烷烃～烯烃

5, 6元环— 稳定

环烷烃反应活性：

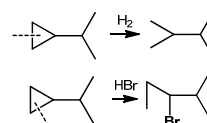
C3 环 > C4 环 > C5-7 环

小似烯，大似烷，高锰酸钾不破坏

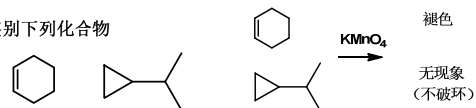
7

测试

完成下列反应

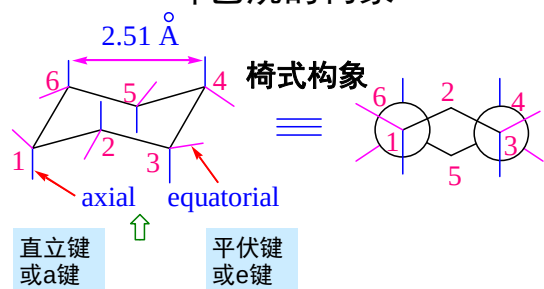


鉴别下列化合物



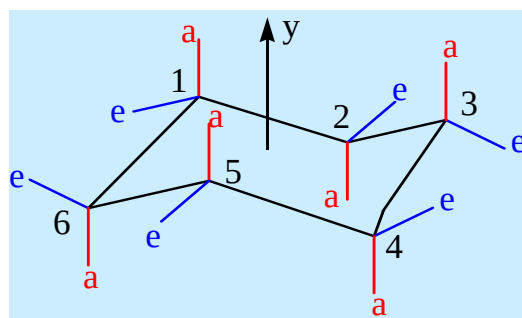
环己烷及其衍生物的构象

环己烷的构象



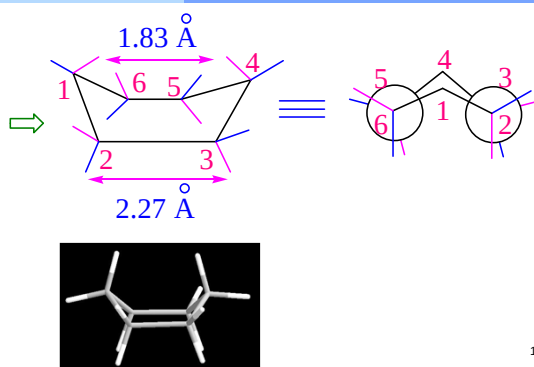
9

直立键和平伏键



10

船式构象



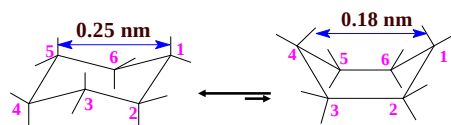
11

以椅式构象为主

根据热力学计算结果推测，在 25℃时，环己烷基本上是以椅型构象存在，船型构象仅约占0.1%

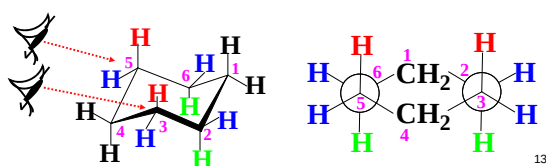
12

椅式构象和船式构象可以相互转变，
但椅式构象比船式构象稳定：



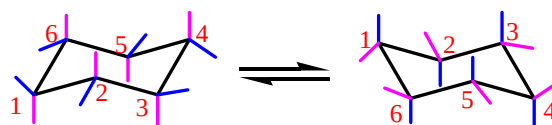
椅式构象稳定的原因：

所有的键处于交叉式



13

椅式构象的翻转

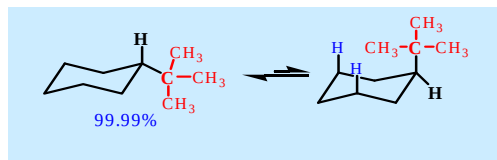
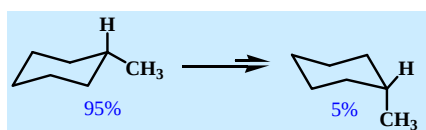


构象转换体

两个椅式构象之间的能垒为44.3 kJ/mol

14

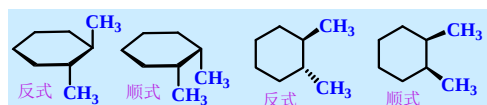
优势构象：体积大的取代基处于平伏键（e键）



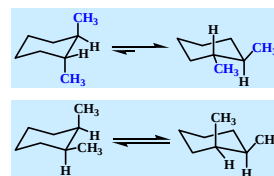
15

二取代环己烷的构象

• 平面式表示法

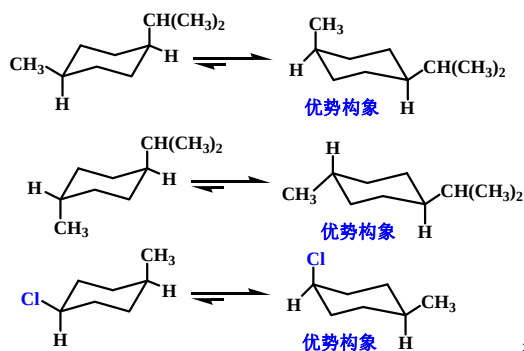


• 构象式表示法



16

两个不同取代基的环己烷衍生物



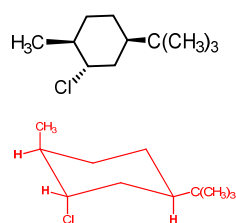
优势构象的判定

- 根据势能差计算
- 含相同基团的多取代环己烷，如果没有其它因素的影响，较多取代基处于e-键的构象为优势构象
- 含不同取代基的多取代环己烷，如果没有其它因素的影响，体积较大基团处于e-键的构象为优势构象

18

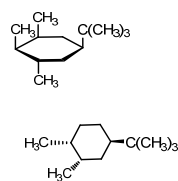
测试

写出下列化合物最稳定椅式构象

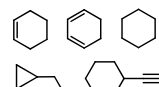


作业

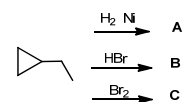
1, 写出下列化合物最稳定椅式构象



2, 鉴别下列五个化合物



3, 完成下列反应



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